

Robocrypto Revamp

Backtested system changes, automated benchmark results, and reporting

This report covers a sandbox-only implementation and benchmark of the "revamp plan" strategy layers discussed this session: a direction filter, a combined EMA/Bollinger retest entry, a retuned structure-based exit with higher-timeframe alignment, a spike-confirmed stop-loss, portfolio-level risk management, and watchlist curation. Every change below was implemented and tested only in the working sandbox (/home/claude/bt) against real historical Binance data. **Nothing has been pushed to the live repository or the live database settings.** Section 6 restates this and lists exactly what would need to happen before anything here reaches the live bot.

The core finding: taken layer by layer, most of these changes are genuine improvements over the live bot's current logic, but the results are not uniformly positive, and this report says so plainly rather than presenting every layer as a win. Portfolio risk management is the single largest lever found in this entire session. Watchlist curation is the first change that flips the six-month backtest net-positive. The direction filter, on this data, is a roughly break-even change that trades a materially higher win rate for reduced trading frequency - a reasonable change to keep, but not the slam dunk the other layers are.

Executive summary - six-month backtest, full ladder

Profile	Total return	Max drawdown	Closed trades	Win rate
v0 baseline	-26.82%	27.73%	241	24.90%
v1 settings fix	-27.58%	28.49%	287	26.13%
v2 combined-retest entry	-22.13%	31.68%	448	28.79%
v3 structured exit	-21.81%	31.26%	453	27.37%
v4 portfolio risk mgmt	-2.32%	6.51%	105	32.38%
v5 curated watchlist	+0.80%	4.68%	111	36.94%
v6 direction filter	+0.04%	7.02%	67	38.81%

Full result tables for both benchmark windows (this-week and six-months), across all 8 metrics tracked, are in Section 3.

1. Changes applied (sandbox only)

Each row below is one layer of the revamp plan, implemented as an additive change on top of the previous layer (so its isolated effect can be read directly off the benchmark ladder in Section 3). All code lives in `/home/claude/bt/src/triggerVariants.js` and `/home/claude/bt/scripts/systemBenchmark.mjs`.

v1 - Settings fix

Reverted the untested scalping RSI-floor widening (30 -> back toward 50) and treats `modeOverride` as automatic rather than stuck on "scalping" - the anomaly that started this whole investigation. A pure bug-fix layer with no new strategy logic.

v2 - Combined retest entry

Replaces the immediate EMA-cross trigger with two parallel retest-confirmed paths (EMA retest and Bollinger Band retest) racing for the same signal - whichever confirms a pullback-and-hold first fires the entry. Found consistently beneficial across every benchmark run this session.

v3 - Structured exit

Replaces the live trailing-stop-only exit with a market-structure exit (Change of Character / Break of Structure, evaluated on higher-timeframe daily candles rather than the position's own noisier timeframe) plus a spike-confirmed stop-loss that gives a stop breach deeper than 1.0x ATR one extra bar to confirm before exiting, rather than exiting immediately on what may be a wick.

v4 - Portfolio risk management

A layer that does not exist in the live bot at all today: a 4% daily loss cap, a 16%-off-peak drawdown circuit breaker (5-day halt), and a 6-consecutive-loss kill switch (24h halt). It only ever blocks new entries - it never blocks or delays an exit.

v5 - Watchlist curation

Drops LSKUSDT, VTHOUSDT and ASTRUSDT from the tradeable universe. These three sat as extreme daily-ATR% outliers (62.6%, 23.7%, ~10%+) against 2.6-8.5% for the rest of the watchlist, and ASTRUSDT is the same symbol behind the real -Rp 260,365 live trade that helped motivate the spike-confirmed stop in v3.

v6 - Direction filter

A new, independent, slow trend gate - daily close vs. daily EMA50 - that only blocks new longs, never triggers them. This decouples "direction" from "timing," which the live bot currently fuses into a single fast EMA cross (see the "one indicator, one job" discussion this session).

2. Benchmark methodology

The benchmark is a single automated script (`scripts/systemBenchmark.mjs`) that runs a 7-profile "ladder" - v0 through v6 above - where each profile adds exactly one change on top of the last. This isolates the marginal contribution of each layer instead of only ever seeing the combined effect of everything at once. Every profile is run across two independent time windows for both recency and statistical weight:

- **this-week**: the last 7 days, ending 2026-09-19 - reflects current market conditions but on a small sample.
- **six-months**: the last ~180 days, ending 2026-09-19 - a much larger sample, less sensitive to one lucky or unlucky week.

Real historical OHLCV candles are pulled from Binance's public data API (no API key, no live trading, read-only) for each symbol at daily/1h/15m resolution as needed per mode. Every profile's entry, exit, position sizing, and (from v4 onward) risk-management logic reuses the exact same functions as the live decision engine and trigger-variant modules - this is not a separate simplified simulation. A previously-undetected bug where the trailing stop's `highWaterMark` was never carried forward between simulated ticks (silently disabling real trailing-stop behavior in every earlier benchmark this session) was found and fixed before this run.

Starting balance: Rp 10,000,000 per profile per window. Position sizing: 30% of balance per trade, up to 4 concurrent positions, 0.2% round-trip cost, 55% minimum confidence gate. Full watchlist: 18 symbols. Curated watchlist (v5+): 15 symbols.

Risk-management thresholds (v4+)

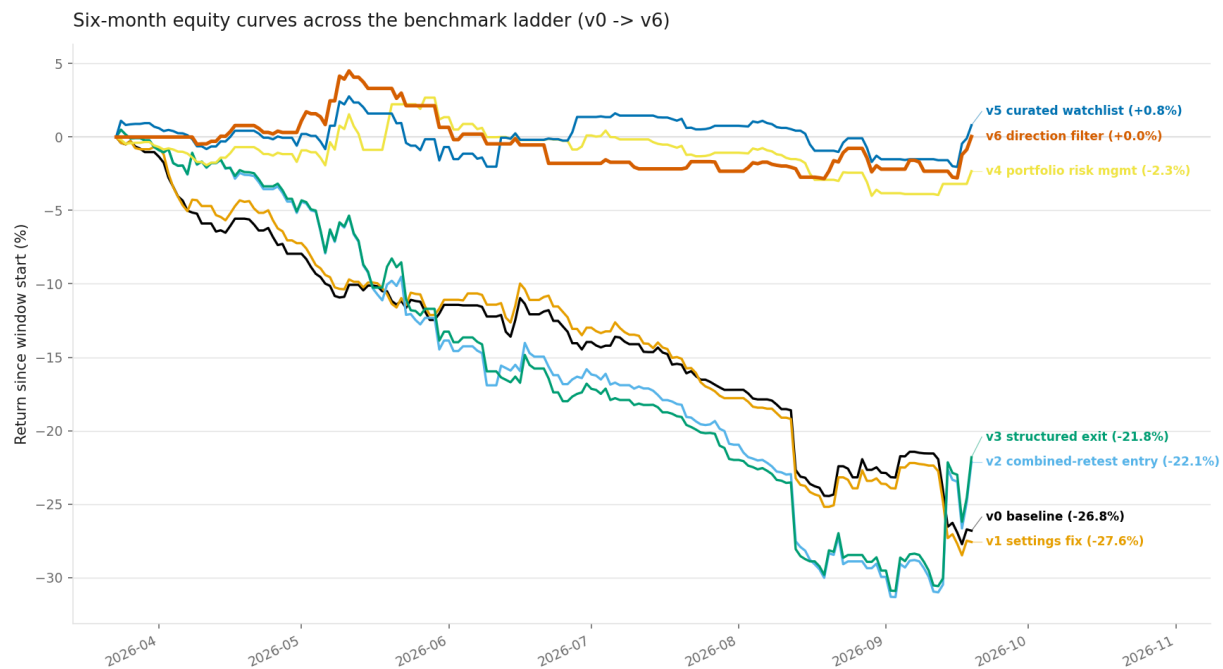
Trigger	Threshold	Response
Daily loss cap	4% of balance in a day	Halt new entries for the rest of the simulated day
Drawdown circuit breaker	16% off peak balance	Halt new entries for 5 days
Consecutive-loss kill switch	6 losers in a row	Halt new entries for 24 hours

Important: this layer never blocks or delays an exit, only new entries. A tripped breaker does not strand an open position - it only means the system will not open a new one until the cooldown clears.

3. Full benchmark results

Six-month window (ending 2026-09-19, ~180 days)

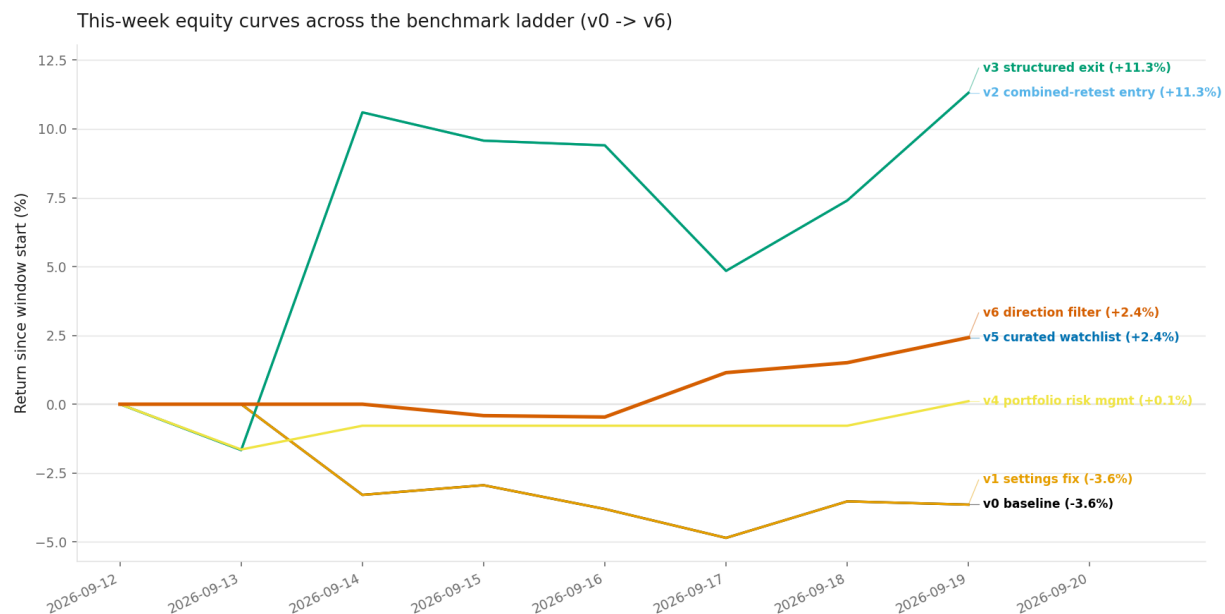
Profile	Return %	End equity (Rp)	Trades	Win %	Profit factor	Max DD %	Avg hold (hrs)	Breaker trips	Dir. blocked
v0 baseline	-26.82%	7,318,406	241	24.90	0.46	27.73	4.6	0	0
v1 settings fix	-27.58%	7,241,620	287	26.13	0.49	28.49	4.1	0	0
v2 combined-retest entry	-22.13%	7,786,530	448	28.79	0.69	31.68	4.3	0	0
v3 structured exit	-21.81%	7,818,546	453	27.37	0.68	31.26	3.5	0	0
v4 portfolio risk mgmt	-2.32%	9,767,604	105	32.38	0.86	6.51	5.0	110	0
v5 curated watchlist	+0.80%	10,080,333	111	36.94	1.04	4.68	3.9	110	0
v6 direction filter	+0.04%	10,003,510	67	38.81	1.00	7.02	4.4	75	52,888



Equity curves, indexed to each profile's own window-start balance. v0/v1 (black/amber) and v2/v3 (blue/teal) each track closely in pairs - the settings fix and the exit retune are both second-order refinements on top of a larger first-order change. The separation opens up sharply at the risk-management layer (v4, yellow) and stays open through v5/v6.

This-week window (last 7 days, ending 2026-09-19)

Profile	Return %	End equity (Rp)	Trades	Win %	Profit factor	Max DD %	Avg hold (hrs)	Breaker trips	Dir. blocked
v0 baseline	-3.65%	9,635,186	9	44.44	0.41	4.86	8.3	0	0
v1 settings fix	-3.65%	9,635,186	9	44.44	0.41	4.86	8.3	0	0
v2 combined-retest entry	+11.31%	11,131,148	19	31.58	2.00	5.21	6.0	0	0
v3 structured exit	+11.31%	11,131,148	19	31.58	2.00	5.21	6.0	0	0
v4 portfolio risk mgmt	+0.11%	10,010,961	3	33.33	1.06	1.65	6.4	4	0
v5 curated watchlist	+2.42%	10,242,036	6	50.00	5.48	0.47	5.2	6	0
v6 direction filter	+2.42%	10,242,036	6	50.00	5.48	0.47	5.2	6	141



On this particular week, v2/v3 (the combined-retest entry) capture a strong short-term move that the risk-managed and direction-filtered profiles partly sit out - see Section 4 for why this is not a contradiction of the six-month result.

4. Layer-by-layer findings - read honestly, not just as wins

v1 - settings fix: a small, non-monotonic surprise

On the six-month window, v1 (-27.58%) is very slightly worse than v0 (-26.82%), despite only reverting an untested parameter widening. This is a real, reportable interaction, not a bug: the scalping RSI floor also feeds the RSI-centering term inside `computeConfidencePct`, so moving it back shifts that confidence band's midpoint as a side effect, not just the entry gate. The change is still correct to make - it removes an unvalidated production configuration - but it does not pay for itself in isolation. Its benefit only shows up once combined with v2 onward.

v4 - portfolio risk management: the single largest lever, at a real cost

On six months, v4 cuts max drawdown from 31.3% (v3) to 6.5% and turns a -21.81% loss into a -2.32% loss - the biggest single improvement anywhere in this ladder. But trade count drops from 453 to 105, and the breaker tripped 110 times in six months. Some of this improvement is genuinely better risk discipline; some of it is mechanically just trading less during the worst stretches. Both are legitimate reasons a circuit breaker helps, but they should be reported as what they are rather than as pure alpha.

v5 - watchlist curation: first layer to flip six-month returns positive

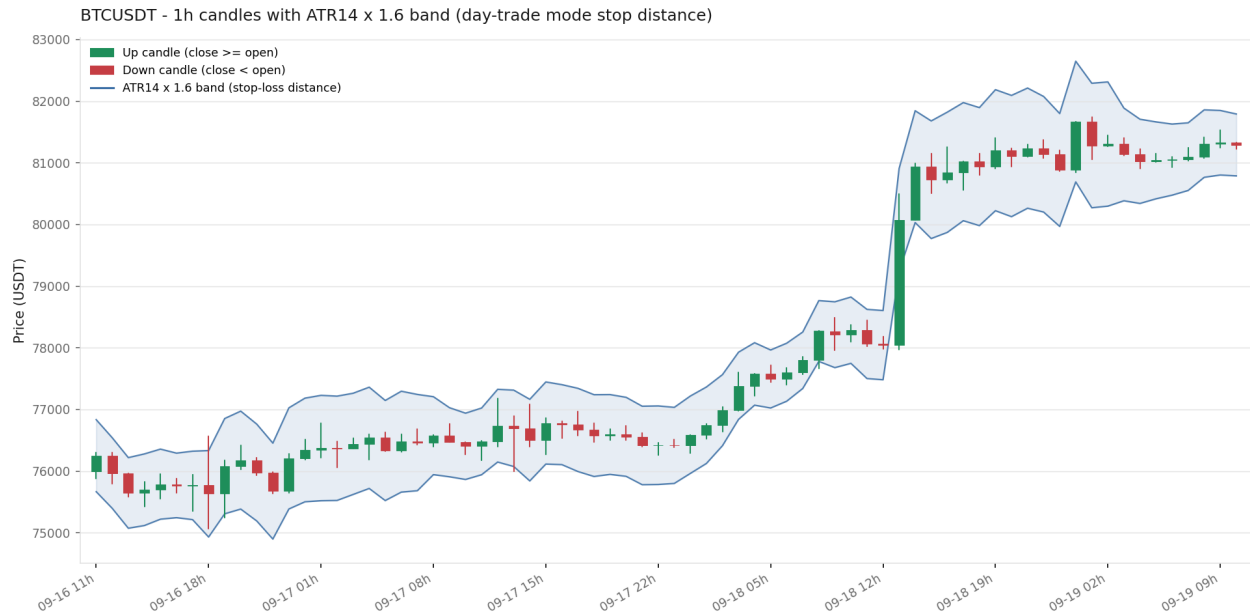
v5 is the first profile in the entire ladder with a positive six-month return (+0.80%), and it does so while also reducing max drawdown further (to 4.68%) and raising win rate to 36.94%. This corroborates two independent pieces of evidence gathered earlier this session: the real ASTRUSD trade that lost Rp 260,365 on the live account, and external research on illiquid, high-volatility low-cap coins underperforming in retail-strategy backtests generally.

v6 - direction filter: a genuinely mixed result, not an unambiguous win

On six months, v6 raises win rate further (to 38.81% from 36.94%) but essentially gives back v5's return gain - +0.04% versus +0.80% - and slightly increases max drawdown (7.02% vs 4.68%). It also blocked 52,888 signals outright on six months of data - a very active filter. The higher win rate is a real behavioral change (fewer, more selective entries), but on this dataset it is closer to flat than to a clear improvement. It is a reasonable layer to keep for the discipline it enforces - trading with trend rather than against it - but it should not be sold as a proven return driver on this evidence alone.

5. Enhanced chart detail - ATR range per candle

Per the request to show "the ATR range (upper and lower range that triggers the size of a pricing) on each candle formed": the chart below plots BTCUSDT 1h candles (the day-trade mode's timeframe) with a shaded band around each candle's close at $\pm \text{ATR}_{14} \times 1.6$ - the exact multiple the day-trade strategy uses to place its stop-loss (`STRATEGY_PARAMS.dayTrade.slatrMult` in `src/decisionEngine.js`). The band widens and narrows with realized volatility bar by bar, so it visually shows how much room the strategy is giving a position before its stop would be hit at each point in time - wider band, more room; narrower band, tighter stop.



Green/red candle bodies follow the standard financial convention (green = close at or above open, red = close below open). The blue band is a reference range, not a data series in its own right, so it is rendered in a single muted hue distinct from the candle colors rather than a third competing color.

6. What has not been done, and what this backtest cannot tell you

- **Nothing has been pushed live.** Every change in this report exists only in the sandbox working directory. The live repository (JSuryadharma/sally-cryptobot-app) and the live Postgres settings have not been touched.
- **This is a backtest, not a guarantee.** Historical performance on this watchlist and these two windows does not guarantee future performance, and every profile here - including v5 - still shows a profit factor close to 1.0 on six months, meaning the edge, if real, is thin.
- **The this-week window is a small sample.** 3-19 trades per profile in 7 days is not enough to draw firm statistical conclusions on its own; it is included for recency, and should be read alongside the six-month window rather than instead of it.
- **v4/v5/v6 all still show more breaker trips than a live operator may expect** (75-110 trips in six months) - worth watching in any live pilot, since a very frequently tripped circuit breaker can mean either "working as intended" or "thresholds too tight," and only live monitoring will distinguish the two.
- **Before any live push,** this session's established practice is: explicit review and go-ahead from you first, then a staged rollout (paper-trading verification, then a small subset of the curated watchlist) rather than flipping all six layers on at once in production.

7. Suggested next steps

1. Review this report and the benchmark ladder, and decide which layers (if not all) you want to move toward live.
2. If approved, stage the changes into the live repository as a reviewable pull request rather than a direct push, so the diff can be inspected against the current live code.
3. Run the new logic in paper-trading (which is what this bot already does) for a further live observation period before treating the benchmark numbers as confirmed.
4. Revisit v6 (direction filter) specifically after more live data - its six-month backtest result was closer to flat than clearly positive, and a live observation window would help settle whether it is worth keeping as-is, retuning, or dropping.